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Effects of elevated pH on marine copepods in mass cultivation systems: practical implications

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Female tolerance to pH (8.0–9.5) by six marine copepods, *Oithona similis*, *Temora longicornis*, *Acartia* spp., *Centropages typicus*, *Pseudocalanus elongatus* and *Eurytemora affinis* was investigated to identify robust species for live feed production. The species with the most oceanic-neritic distribution, *O. similis*, exhibited 72 h LC₅₀ at pH 8.39 ± 0.11 (±95% CL) whereas the most estuarine *E. affinis* had LC₅₀ at pH 9.51 ± 0.04. The rest had LC₅₀ at intermediary pH's. Egg hatching by a selection of species, *Acartia* spp., *C. typicus* and *E. affinis*, was unaffected by pH up to 9.0–9.5. Nauplii from both *Acartia* spp. and *C. typicus* had higher mortality at pH 9.5 than at the other pH regimes while *E. affinis* nauplii were not affected by pH. Wild *Acartia* spp. and *A. tonsa* from a culture showed some differences in response although of minor practical importance for aquaculture; both produced no eggs at pH 9.5, *A. tonsa* exhibited significantly higher egg production at all other pH's than 9.5, both showed egg hatching invariant of pH, but gradually increasing nauplii mortality with pH. We suggest active/passive selection to obtain the most pH robust species able to cope with accidentally, but frequently, elevated pH in aquaculture systems.

KEYWORDS: copepods; elevated pH; egg production and hatching; adult and nauplii survival; live feed; aquaculture

INTRODUCTION

Larviculture of most marine species requires live feed. It has been shown that copepods are a superior diet for many fish larvae due to their size, swimming behaviour,

and their biochemical profile (e.g. Buskey, 2005; Turingan *et al.*, 2005; Drillet *et al.*, 2006). In semi-intensive outdoor copepod cultures, it is imperative to understand pH effects on copepod assemblages in order to eventually

select for robust species able to resist large fluctuations in environmental parameters. Productive cultures are characterized by high photosynthesis in which the zooplankton are at risk of exposure to pH levels exceeding 9.5 in e.g. nutrient amended open tanks (Blanda *et al.*, 2015a). At severely elevated pH levels, the copepod assemblage has been reported to change from a ~50% dominance of *C. hamatus* towards >90% *Acartia* spp. (see for a full report in Blanda *et al.*, 2015a and Jakobsen *et al.*, 2015). In a similar hypereutrophic (24–177 µg Chl a L⁻¹) tropical Taiwanese pond system, seeded by estuarine water and copepods, pH was 8.7 ± 0.1 and thereby elevated by approximately 0.5 units (Blanda *et al.*, 2015b). This system hosted in total 14 mesozooplankton taxa. However, the assemblage was skewed towards 30% and very often $\geq 80\%$ *Pseudodiaptomus annandalei* and the rest dominated by *Apocyclops royi*. These two examples suggest a general species selection from a species rich inoculum to a few tolerant species when pH is driven up by high primary productivity. Hence, we anticipate that pH will impact the composition of copepod assemblages in semi-intensive copepod rearing tanks.

Continuous laboratory copepod cultures are subjected to strong selection, as opposed to natural- and semi-intensive tank populations, since they are kept at relatively constant abiotic conditions (temperature, pH and often a 24 h light cycle), excess food and no predators (*sensu* Hammervold *et al.*, 2015). This might impact various physiological traits of the copepods and potentially means that laboratory culture specimens differ in physiological responses from the wild population (Tiselius *et al.*, 1995). Here, we compare adult survival, egg production, egg hatching and nauplii survival of *Acartia* spp. from a wild population vs. *A. tonsa* from our continuous culture when exposed to elevated pH. We realize that a direct comparison of effects on a species level is not possible since the wild *Acartia* population is an unidentified assemblage, but here we take the approach of comparing pH effects on *Acartia* at a genus level. Our motivation for conducting this particular part of our study is that we consider pH tolerance in all copepod species, and their life stages, that are relevant to aquaculture as crucial knowledge for understanding performance of copepods in all types of aquaculture rearing facilities. With that in mind, we strive to investigate the possibility for selection of pH robust copepod species to circumvent one of the most obvious rearing limitations while optimizing management procedures in the live feed aquaculture industry. As elevated pH can also affect inorganic nutrient dynamics, that are potentially toxic for copepod cultures (Buttino, 1994; Jepsen *et al.*, 2015), addressing pH tolerance is of vital importance for the aquaculture industry. We hypothesize, based on Mauchline's (1998) copepod habitat occupation

definitions, that (1) neritic-estuarine copepod species generally exhibit high tolerance to elevated pH, (2) that the most oceanic-neritic copepod species is less tolerant to elevated pH than the most pronounced estuarine copepod species and (3) that the *Acartia tonsa* specimens from our >30 year old continuous mono species culture exhibit less tolerance towards elevated pH than a wild *Acartia* spp. assemblage collected from inner Danish waters and reared in a semi-intensive outdoor rearing system.

Here, we tested the pH tolerance of some of the copepod species present in the Danish mass culture system described by Blanda *et al.* (2015a) besides some other common Baltic Sea coastal species within a pH gradient of 8.0–9.5. This was done through laboratory exposure of pelagic copepods by acute tests. We used adult female survival for all species, egg production and hatching, and nauplii survival as end-points for a selection of species. We discuss the practical implications of elevated pH for aquaculture rearing of boreal neritic calanoids.

METHOD

Copepod sampling, culture and experimental conditions

Six species of marine copepods were selected for the laboratory experiments. These copepods were chosen based on their availability and that they represent a natural gradient from oceanic-neritic (*Oithona similis*) towards more estuarine habitats (*Temora longicornis*, *Acartia* spp., *Centropages typicus*, *Pseudocalanus elongatus*) and extremely low salinity to freshwater habitats (*Eurytemora affinis*) according to OBIS (OBIS/UNESCO, 2016). Eutrophication is typically found in low saline coastal regions. This salinity gradient is typically correlated with eutrophic regions and is also likely to exhibit a gradual increasing fluctuation in pH particularly elevated pH. Some species originated from cultures, while others were sampled during summer of 2004 in Danish coastal waters (for details see Table I) using a WP-2 plankton net with a mesh size of 50 or 250 µm. Copepods from the closed cod-end were gently diluted in surface seawater in a 70 L insulated box immediately after retrieval and then transferred to the laboratory within a few hours. All wild and cultured copepod species were kept in 70 L tanks at salinity 30, temperature of 17°C, pH 8.2 and dim light (5 µmol photons m⁻² s⁻¹). The animals were fed the cryptophyte *Rhodomonas salina* (strain SCCAP K-1487) in excess (50 000 cells ml⁻¹ equal to 2370 µg C L⁻¹) according to Berggreen *et al.* (1988) until experimental use (2–10 d later).

The pH was elevated stepwise by pipette additions of 1 M NaOH and finally adjusted using NaOH and HCl.

Table I: Copepods used in the different laboratory experiments

Species	Origin	Culture/wild	Experiment type (adult survival, egg production, egg hatching, nauplii survival)	Reference
<i>Oithona similis</i>	Øresund, DK	Wild	Adult survival	By authors
<i>Temora longicornis</i>	Øresund, DK	Wild	Adult survival	By authors
<i>Acartia</i> spp.	Øresund, DK	Wild	Adult survival, egg production, egg hatching, nauplii survival	By authors
<i>Acartia tonsa</i>	Øresund, DK	Culture strain DFH-AT1	Adult survival, egg production, egg hatching, nauplii survival	Støttrup <i>et al.</i> (1986)
<i>Centropages typicus</i>	Øresund, DK	Culture	Adult survival, egg hatching, nauplii survival	Thomas Kjørboe
<i>Pseudocalanus elongatus</i>	Øresund, DK	Culture	Adult survival	Espen Bagøyen
<i>Eurytemora affinis</i>	Randers Fjord, DK	Wild	Adult survival, egg hatching, nauplii survival	By authors

The pH measurements were carried out using a Sentron pH metre equipped with a red-line electrode, sensitivity 0.01, using a 2-point calibration. To minimize potential shock effects of high pH on the organisms, the pH was raised by slowly pipetting NaOH to levels of 0.5 unit increments at 12 h intervals until the final value was reached with the exception of pH 8.0–8.5 which was adjusted immediately. Hence, the entire acclimation process lasted for 24 h no matter the treatment after the adjustments were initiated.

Female survival experiments on six species

To monitor the adults' response, the effect of high pH on the survival of females from the six species of copepods was initiated by acclimation to the different pH levels. Between 25 and 40 females were pre-acclimated by incubation in acid washed Nalgene bottles (2.8 L) at four levels of pH: 8.0, 8.5, 9.0 and 9.5 and adjusted as described above while presented with excess food algae. After the 24 h pre-acclimation period, five live animals (egg sac carriers were deliberately chosen without egg sacs) were individually picked with a Pasteur pipette and added to Nalgene bottles (600 ml) in a maximum of five bottles at each pH level (five replicate bottles with five females in each). These bottles contained *R. salina* in excess (*sensu* Berggreen *et al.*, 1988). Due to mortality, especially in the high pH acclimation treatments, it was however not always possible to conduct five replications with five individuals in 600 ml bottles (Table II). When below 15 females in total (three replicates of five individuals in 600 ml) we chose to incubate individual females in 65 ml Nalgene bottles. Both 600 ml and 65 ml incubation bottles were mounted on a plankton wheel (1 rpm) and incubated for 3 days at $17 \pm 1^\circ\text{C}$ on a 16:8 h light:dark cycle at an irradiance of $5 \mu\text{mol photons m}^{-2} \text{s}^{-1}$. The animals were inspected daily in the bottles under a dissecting microscope and considered dead when lying on the bottom of the bottles and no movement was observed for 1 min.

Egg hatching experiments and nauplii mortality in three copepod species

To monitor effects of elevated pH on copepod recruitment, egg hatching experiments were performed from batches of *Acartia* spp., *C. typicus* and *E. affinis* eggs at four levels of pH: 8.0, 8.5, 9.0 and 9.5. For each species, 3–4 replicate Petri dishes with seawater holding a registered number of >12 eggs each (Table II) were incubated. Eggs from the egg-carrying *E. affinis* were detached from the females before incubation. After 48 h incubation, petri dishes were inspected for the number of eggs, dead nauplii and swimming nauplii. Hatching success and nauplii mortality was calculated.

Female survival, egg production, egg hatching and nauplii mortality of *Acartia* spp. from the wild vs. *A. tonsa* from a culture

Female *Acartia* (1–6 females in six replicates, Table II) were sorted and thereafter incubated in 600 ml acid rinsed NUNC polycarbonate bottles added 40 000 *R. salina* cells ml^{-1} equal to $1896 \mu\text{g C L}^{-1}$. The females were transferred daily to new suspensions (pH: 8.0, 8.5, 9.0 and 9.5 and food adjusted) and their egg production monitored after 48 h. Experiments on egg hatching were thereafter performed with both *Acartia* populations at the same four levels of pH. For each *Acartia* population, 3–4 replicate Petri dishes with filtered Øresund seawater and >30 enumerated eggs in each dish (Table II) were incubated. After 48 h of incubation, the petri dishes were inspected for the number of eggs, dead nauplii and swimming nauplii. Hatching success and nauplii mortality were calculated.

Statistics

The model used to analyse the mortality effects on female copepods was fitted with 'Log (agonist) vs. response – Find EC anything'. The Top and Bottom plateau's have been constrained to 0 and 100, since the mortality ranges from

0% to 100% and $LC_{50} \pm 95\%$ CL extracted (GraphPad Prism version 5.04, for model details see Table III and IV). The different pH tolerances can be judged as statistically different by comparing LC_{50} values and their respective 95% CL within and across pH tolerances. For the rest of the results in the present study, statistical analyses were performed in Sigmaplot for Windows version 10.0 build 10.0.0.54. Percentage data (e.g. hatching success, mortality) were arcsin transformed before statistical analyses

were performed. One-way ANOVA and the post hoc Holm-Sidak test were used to evaluate the within species effect of pH on egg hatching success of *Acartia* spp., *A. tonsa* and *C. typicus*. Data from *E. affinis* was not normally distributed and therefore an ANOVA on ranks and the post hoc Kruskal–Wallis test was applied. One-way ANOVA and the post hoc Holm-Sidak test was also used to evaluate the effect of pH on nauplii mortality of *Acartia* spp., *A. tonsa*, *C. typicus* and *E. affinis*.

Table II: An overview of the conducted experiments

Experiment	#Females exposed in each replicate 600 ml bottles in pH gradient	#Eggs initially in each replicate exposed in pH gradient	#Replicates
Female survival on six species of copepods	<i>Oithona similis</i> 4–5 <i>Temora longicornis</i> 4–5 <i>Acartia</i> spp. 2–5 <i>Acartia tonsa</i> 3–6 <i>Centropages typicus</i> 2–5 <i>Pseudocalanus elongatus</i> 4–6 <i>Eurytemora affinis</i> 2–6		3–5 (600 ml) Or if individually incubated 10–14 (65 ml)
Egg hatching and nauplii mortality experiments on three copepod species		<i>Acartia</i> spp. 30–128 <i>Centropages typicus</i> 12–46 <i>Eurytemora affinis</i> 17–32	3–4
Egg production by <i>Acartia</i> spp. and <i>A. tonsa</i>	<i>Acartia</i> spp. 1–5 females per replicate; 0–128 eggs produced per replicate <i>Acartia tonsa</i> 2–6 females per replicate; 3–199 eggs produced per replicate		6
Egg hatching and nauplii mortality by <i>Acartia</i> spp. and <i>A. tonsa</i>		<i>Acartia</i> spp. 30–128 <i>Acartia tonsa</i> 30–151	3–4

Table III: Wild adult female copepods pH survival response given as pH level where 50% mortality was observed (LC_{50} – lethal time) after 24, 48 and 72 h incubations, see text

Time (h)	LC_{50}	95% CL	Hill slope	SE	df	SS	r^2	Species
24	~8.74	very wide	N/A	N/A	18	0	1.00	<i>O. similis</i>
48	8.60	(8.04–9.17)	–6.77	17.63	18	1579	0.96	<i>O. similis</i>
72	8.39	(8.28–8.50)	–1.61	0.29	18	2981	0.87	<i>O. similis</i>
24	N/A	N/A	N/A	N/A	32	0	1.00	<i>T. longicornis</i>
48	8.83	(8.69–8.97)	–1.41	0.28	32	12 832	0.74	<i>T. longicornis</i>
72	8.44	(8.38–8.51)	–2.16	0.36	32	5494	0.89	<i>T. longicornis</i>
24	N/A	N/A	N/A	N/A	22	0	1.00	<i>Acartia</i> spp.
48	9.03	(8.88–9.19)	–1.93	0.62	22	11 098	0.73	<i>Acartia</i> spp.
72	8.72	(8.43–9.02)	–1.1	0.39	22	21 651	0.46	<i>Acartia</i> spp.
24	N/A	N/A	N/A	N/A	25	0	1.00	<i>C. typicus</i>
48	9.31	(9.16–9.47)	–1.9	0.55	25	12 227	0.49	<i>C. typicus</i>
72	8.90	(8.70–9.11)	–1.48	0.47	25	19 339	0.57	<i>C. typicus</i>
24	N/A	N/A	N/A	N/A	22	0	1.00	<i>P. elongatus</i>
48	9.65	(9.47–9.84)	–1.81	0.71	22	2903	0.63	<i>P. elongatus</i>
72	9.24	(9.18–9.30)	–2.99	0.36	22	4928	0.94	<i>P. elongatus</i>
24	N/A	N/A	N/A	N/A	18	0	1.00	<i>E. affinis</i>
48	N/A	N/A	N/A	N/A	18	0	1.00	<i>E. affinis</i>
72	9.51	(8.47–9.54)	–0.38	0.24	18	13 955	0.13	<i>E. affinis</i>

df are $n-2$ used in the model fit.

RESULTS

Female survival experiments on six species

In our pH exposure experiments, practically none of the species, except for female *O. similis*, exhibited mortality after 24 h of incubation. In fact, *O. similis* died out somewhere between pH 8.5 and 9 (Fig. 1). The results from 48 and 72 h incubations indicated that *O. similis* was generally more vulnerable to elevated pH than the other species while *E. affinis* was the most tolerant. In Table III, $LC_{50} \pm 95\%$ CL is listed for 48 and 72 h exposures. It is shown that the mean LC_{50} pH values for all species was in the range of 8.60–9.65 and declined

with incubation time. Moreover, it is clear that *O. similis* is the least tolerant (pH 8.60) and *E. affinis* the most tolerant (pH 9.51), approximately an entire pH unit more tolerant after 72 h, and the pH tolerance of the rest of the copepod species are positioned in between.

Egg hatching experiments and nauplii mortality on three copepod species

Egg hatching success vs. pH up to 9.0 varied among species, with values from 100% (*E. affinis*) regardless of pH, >80% (*Acartia* spp.) regardless of pH, and <20 to >40% (*C. typicus*) (Fig. 2). Surprisingly, pH did not have

Table IV: Wild adult female *Acartia* spp. and cultured *Acartia tonsa* copepods pH survival response after 24, 48 and 72 h incubations

Time (h)	LC_{50}	95% CL	Hill slope	SE	df	SS	r^2	Species
24	N/A	N/A	N/A	N/A	22	0	1.0	<i>Acartia</i> spp.
48	9.03	(8.88–9.19)	–1.93	0.62	22	11 098	0.73	<i>Acartia</i> spp.
72	8.72	(8.43–9.02)	–1.10	0.39	22	21 651	0.46	<i>Acartia</i> spp.
24	N/A	N/A	N/A	N/A	18	0	1.0	<i>A. tonsa</i>
48	9.49	(9.42–9.55)	–3.4	2.26	18	2802	0.78	<i>A. tonsa</i>
72	8.99	(8.90–9.07)	–4.27	6.91	18	6193	0.83	<i>A. tonsa</i>

df are $n-2$ used in the model fit.

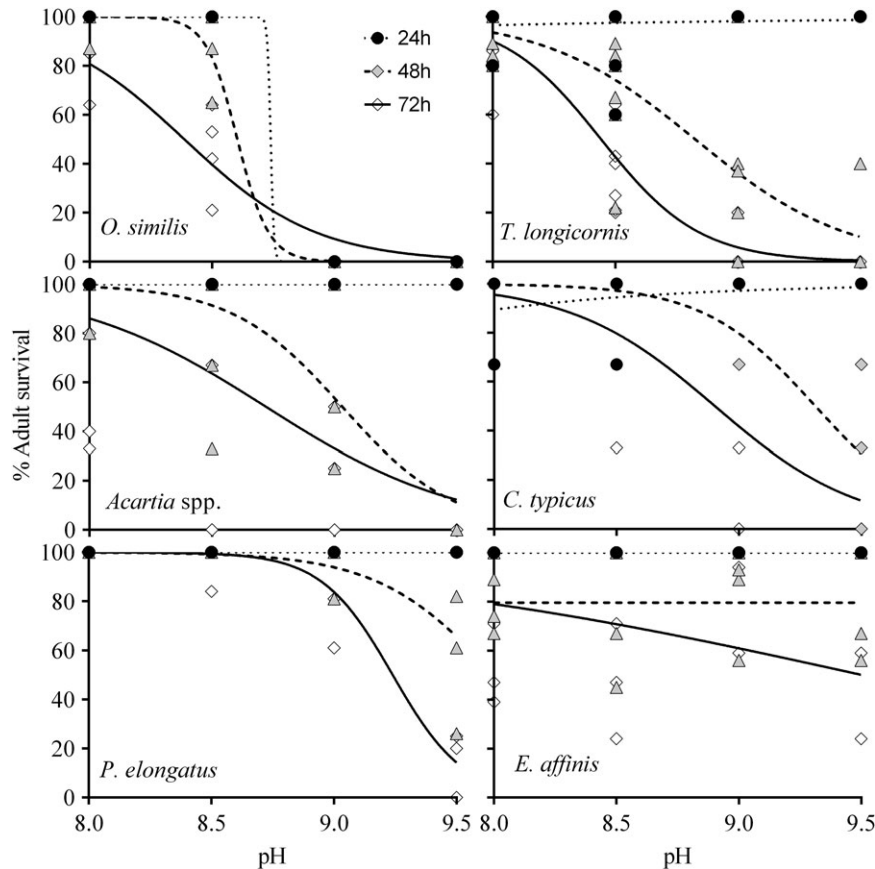


Fig. 1. Wild adult female copepods survival vs. pH levels of the six target species.

any effect on hatching success for either *Acartia* spp. ($df = 16$, $P = 0.138$), *C. typicus* ($df = 10$, $P = 0.733$) or *E. affinis* ($h = 0.000$, $df = 3$, $P = 1.000$). Nauplii from

both *Acartia* spp. ($df = 16$, $P < 0.001$) as well as *C. typicus* ($df = 13$, $P < 0.001$) had significantly higher mortalities at pH 9.5 compared to the other pH regimes. *E. affinis* nauplii mortality was not at all affected by pH ($df = 11$, $P = 0.122$) (Fig. 2).

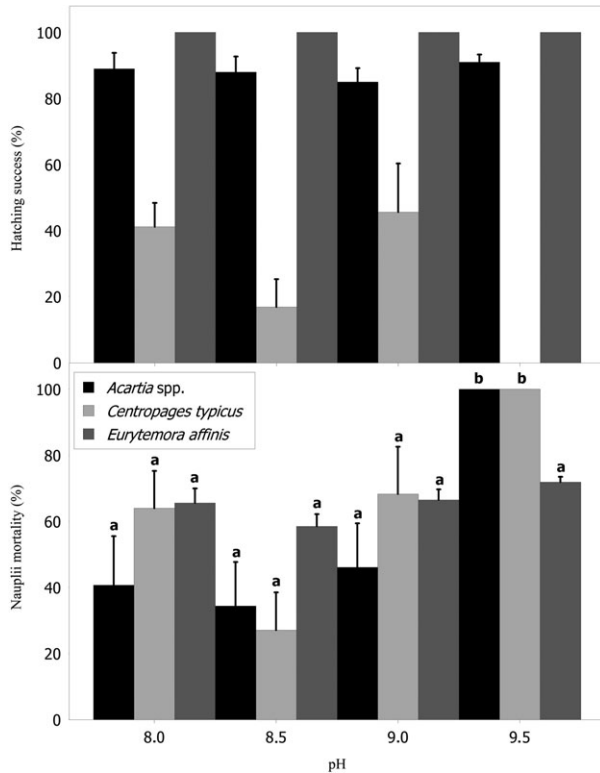


Fig. 2. Egg hatching and nauplii mortality (d^{-1}) of the wild copepods *Acartia* spp., *Centropages typicus*, and *Eurytemora affinis* vs. pH levels. The annotated **a** and **b** are statistically significant different from each other.

Female survival, egg production, egg hatching and nauplii mortality of *Acartia* spp. from the wild vs. *A. tonsa* from a culture

A comparison between responses to elevated pH by wild adult *Acartia* spp. and *A. tonsa* from the culture revealed minor differences in response patterns by the two populations (Fig. 3); however, the survival responses for the adult copepods were significantly different after 48 h exposure to elevated pH, with *Acartia* spp. being affected at slightly lower levels of pH than *A. tonsa* (overlapping 95% CL in Table IV). The results on egg production revealed no eggs produced at pH 9.5 for both populations. At all pH levels lower than 9.5, significantly higher egg production was observed for *A. tonsa* when compared to *Acartia* spp. Egg hatching success between populations at pH 8.0 and pH 9.5 was different as the cultured animals revealed significantly lower hatching success at 9.5 than the wild *Acartia* (Fig. 4 and Table VA). Nauplii mortality increased gradually up to 100% at pH 9.5 for both *Acartia* populations (Fig. 5). Moreover, significantly different nauplii mortalities were found between species at pH 8. Within *Acartia* spp., nauplii mortality at pH 9.5 was significantly different and higher than at all other pH treatments. Within *Acartia tonsa*, nauplii mortality at pH 9.0 and 9.5 was

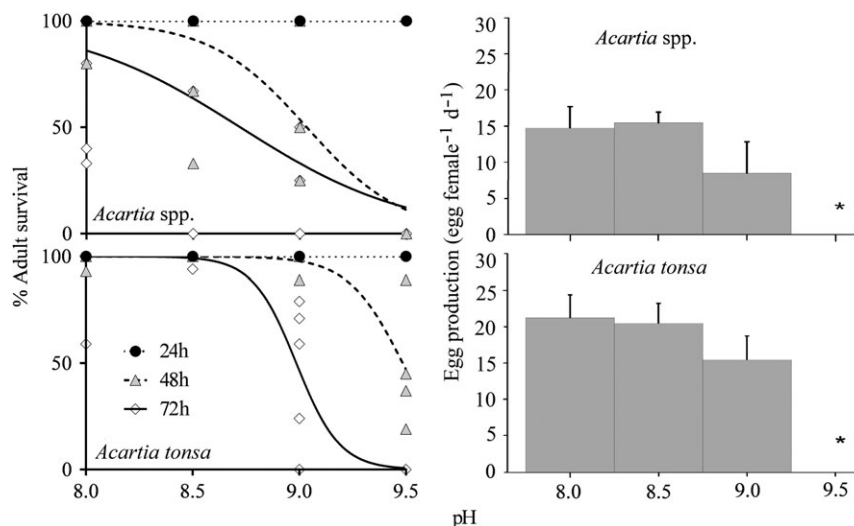


Fig. 3. Adult copepod survival (%) and 24 h and mean $\pm$ SD egg production after 48 h incubation (d^{-1}) vs. pH levels by wild *Acartia* spp. and cultured *A. tonsa*. * indicate zero egg production.

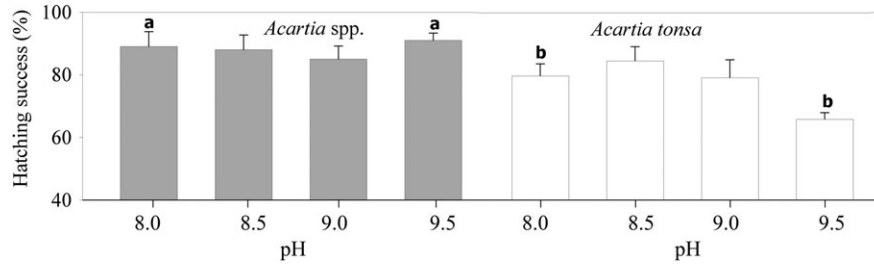


Fig. 4. Comparison of egg hatching success (%) between *Acartia* spp. (grey bars) collected in the wild vs. *A. tonsa* (white bars) originating from laboratory cultures. A statistically significant difference was found between species (*Acartia* spp. vs. *Acartia tonsa*) at pH 8.0 and pH 9.5 (annotated with **a** and **b**). Bars are means and error bars are s.d.

*Table V: (A) Two-way Anova table of hatching success of eggs and (B) two-way Anova table of nauplii mortality of wild *Acartia* spp. vs. cultured *A. tonsa*, respectively*

	df	SS	MS	F	P
(A) Source					
pH	3	0.19	0.0632	1.615	0.207
Species	1	0.481	0.481	12.282	<0.001
pH × species	3	0.261	0.087	2.224	0.106
Residual	30	1.174	0.0391		
(B) Source					
pH	3	7.928	2.643	53.444	<0.001
Species	1	0.0301	0.0301	0.609	0.441
pH × species	3	0.208	0.0693	1.402	0.261
Residual	30	1.483	0.0494		

df are $n-1$.

Significant *P*-values are in bold style.

significantly different and higher than at all other pH treatments (Table VB).

DISCUSSION

Elevated pH in natural habitats and its relevance for aquaculture

It is well known that pH fluctuates in lakes, diurnally and seasonally, depending on alkalinity and photosynthetic activity, and pH as high as 10 has been reported (Talling, 1976). In contrast, pH is quite stable around pH 8.0 ± 0.3 in marine waters (Doney *et al.*, 2009; Duarte *et al.*, 2013). The least seasonal variations are found in the tropical open ocean (<0.1 pH unit; Doney *et al.*, 2009) while the largest variations have been observed in shallow waters in coastal areas, where pH can easily exceed 9 in the photic zone of the water column during algal blooms (e.g. Pedersen and Hansen 2003a; Middelboe and Hansen, 2007). In the Gulf of Finland, Baltic Sea, pH fluctuated between 7.4 and 9.2 during 1972–2009 and the seasonal variation in pH is largely driven by phytoplankton blooms (Brutemark *et al.*, 2011). In hypereutrophic estuaries like e.g. Mariager

Fjord, Denmark, pH can actually reach up to 9.75 at mid-day during high summer (Hansen, 2002). This pH fluctuation, and especially the peak value, is a structuring parameter for the phytoplankton community (Hansen, 2002; Hinga, 2002; Lundholm *et al.*, 2004) and protozoans (Pedersen and Hansen, 2003a). The same phenomenon has been suggested for marine copepods by Pedersen and Hansen (2003b) and Tiselius *et al.* (2008), but hitherto not studied systematically. Semi-intensive rearing systems often mimic natural fluctuations (Jakobsen *et al.*, 2015), particularly when nutrient enriched for higher primary production. These frequently exhibit elevated pH (Blanda *et al.*, 2015a,b) and this is why the perspective of pH structuring copepod assemblages resembles that of *in situ* conditions.

Species specific tolerance to elevated pH and the relation to their habitat

We hypothesized that neritic-estuarine copepod species generally exhibit higher tolerance to elevated pH than the most oceanic-neritic species. We found that the females tested for each species tolerate elevated pH. However, considerable variation in tolerance was observed for the different species. The pH tolerance of the tested species seems linked to their natural distribution. *Oithona similis* is a true oceanic-neritic species and exhibited less tolerance to elevated pH, since a significant increase in mortality appeared already at a slightly elevated pH of 8.5. Oceanic-neritic habitats are considered quite stable in pH (range 7.5–8.5, Duarte *et al.*, 2013), which is why species from such a habitat are expected not to be as tolerant to elevated pH. The more euryhaline species (see Mauchline, 1988) e.g. *T. longicornis*, *Acartia* spp., *C. typicus*, *P. elongatus*, and in particular *E. affinis* are adapted to a very fluctuating environment in estuaries where elevated pH is a common stressor (Hansen, 2002). Females representing the extreme end points of the habitats, being the most oceanic-neritic *O. similis* and the most pronounced estuarine *E. affinis*, actually showed

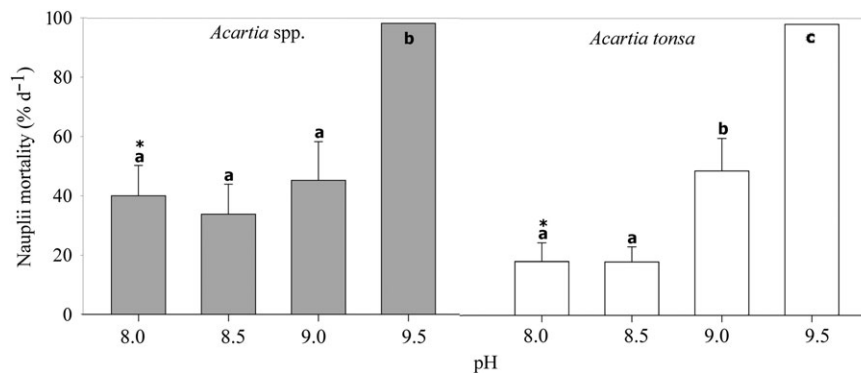


Fig. 5. Comparison of nauplii mortality (% d⁻¹) vs. pH levels between *Acartia* spp. (grey bars) collected in the wild vs. *A. tonsa*. (white bars) originating from laboratory cultures. A statistically significant difference was found between species at pH 8.0 (annotated with *). Within *Acartia* spp. pH 9.5 was statistically significantly different from all other pH treatments (annotated with **a** and **b**). Within *Acartia tonsa*, pH 9.0 and 9.5 was statistically significantly different from all other pH treatments (annotated with **a**, **b** and **c**). Bars are means and error bars are SD.

the least and the most tolerance to elevated pH with approximately one pH unit in difference. Hence, both the first and second hypothesis are accepted.

The most pH tolerant species *E. affinis* is a suitable fish larval feed. However, the main trade-off by using this species is that it is an egg carrying calanoid. It is well known that these have a generally lower specific productivity as compared to free spawning calanoids, like *Acartia* spp. (Hirst and Lampitt, 1998). One can speculate that *Acartia*'s relatively higher productivity measured as a numerical response at excess food most likely outcompete *E. affinis* in artificial systems as in a semi-intensive rearing system (Blanda *et al.*, 2015a) as well in the hypereutrophic estuary Mariager Fjord (Hansen, 2002). In the Fjord, the observed copepod assemblage was qualitatively similar to that of the seeding populations in the Danish semi-intensive mass cultivation system (Blanda *et al.*, 2015a). Only <5% of the total assemblage was anything other than *A. tonsa* in the Fjord. This skewing of the copepod assemblage in Mariager Fjord was most likely due to pH challenges (Tiselius *et al.*, 2008).

Comparison of tolerance to elevated pH by wild *Acartia* spp. assemblage and mono species *A. tonsa* culture

Since *A. tonsa* has shown very promising potential as live feed for fish larvae (Wilcox *et al.*, 2006; Cassiano *et al.*, 2012; Øie *et al.*, 2017) we also tested a wild *Acartia* spp. assemblage vs. a laboratory culture of *A. tonsa*. We found few significant differences in response to elevated pH on adult survival, egg production, hatching and nauplii survival patterns. One interesting observation, however, was that egg hatching success is slightly lower for the cultured *Acartia*, which might be an indicator that when an organism has been out of its natural environment

their eggs lose some resistance towards hatching at elevated pH. The few statistical differences were, for the most of them, in favour of *A. tonsa*, and none of them are expected to have practical implications in a copepod mass production system. Hence, we cannot accept our third hypothesis where we state that the mono species culture is expected to be less robust towards pH elevation. One could have expected general significant genotypic limitations and crucial differences between a naturally occurring assemblage and a long-term cultivated sibling species due to separation from natural fluctuations. In conclusion, the continuously cultivated *A. tonsa* did not respond dramatically differently than the mixed species *Acartia* spp. from the wild and can be considered as a quite valid model system for boreal *Acartia* species concerning response to elevated pH.

The embryo is protected from elevated pH during embryogenesis

One can speculate why egg hatching of *Acartia* spp. at elevated pH is relatively unaffected even at pH 9.5, but the hatched nauplii survival is increasingly poor with gradually elevated pH and close to zero at pH 9.5. This must be ascribed to the same phenomenon as proposed by Nielsen *et al.* (2006) and Hansen *et al.* (2012) when exposed to hydrogen sulphide or salinity challenges and by Grenvald *et al.* (2013) for pyrene exposure on embryos. The embryo is suggested to be protected towards harsh environmental conditions in the egg by an embryonic membrane (Hansen *et al.*, 2012) since they hatch even at very high pH. In fact, we have preliminary results suggesting that hatching takes place even at pH 10 as well as at pH 6 (Jakob Kring Højgaard, Roskilde University, pers. observations). However, when facing the ambient water column environment, an

increasing fraction of the nauplii die right after hatching when pH deviates more than one unit from natural pH. The phenomenon must be an adaptation to harsh environmental conditions in the pelagic at high primary productivity, generating elevated pH, and a protection towards acidic conditions when buried in the sediments due to e.g. hydrogen sulphide. Here, the embryos spend their resting periods in quiescence or diapause and the eggs can occasionally even pass intestinal environments with low pH and survive. This egg membrane system and its physiological protective property have recently been argued to be of potential use for aquaculture (e.g. Hansen *et al.*, 2016). Quiescent eggs are proposed to be included in an egg bank concept, where eggs from tank sediments such as the ones described by Sørensen *et al.* (2007) and Engel-Sørensen *et al.* (2004) are harvested, rinsed and stored in quiescence for later use in the pond manager's copepod rearing programme. The idea is that, when necessary, the tank water could be inoculated with newly hatched nauplii from eggs from the bank originating from production by former copepod generations in the tanks (see details in Hansen *et al.*, 2016).

CONCLUSIONS

We suggest that copepod community composition and succession in Danish eutrophic estuarine systems is driven by elevated pH toward an *Acartia* spp. dominated copepod assemblage. This is interesting basic ecological knowledge on how these estuaries function as well as an important implication for nutrient enriched semi-intensive mass cultivation systems based on boreal copepod assemblages. The latter is crucial since *Acartia* spp. is considered a very good live feed for marine fish hatcheries. Apparently, one does not need to conduct complicated selection efforts aiming at obtaining a more uniform copepod population; it all happens by itself with limited human interference in nutrient enriched rearing tanks driven by a high primary production.

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